Neural Network for Handwritten Digit Recognition Using MNIST

Deep Learning CNN design using MNIST

Key points

Dataset: MNIST

Data Preparation: Getting the data ready for training.

Model Selection: Why choose a Convolutional Neural Network?

Building & Training: Creating and teaching the model

Performance Evaluation: How well did our model do?

Method Comparison: How does this design compare to other methods?

Conclusion & Future Work: Key takeaways and Summary

The Dataset: MNIST

A classic computer vision dataset, often called “Hello World” of ML

70,000 greyscale 28x28 pixel images.

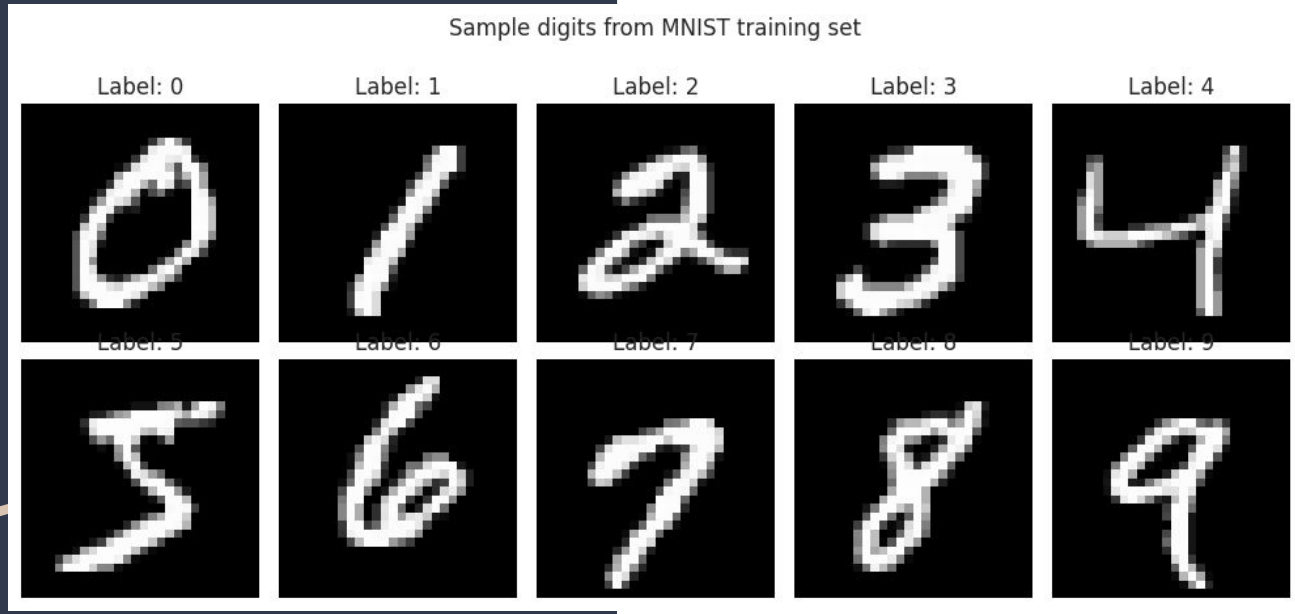
10 Classes: Digits 0 through 9

Pre split: 60,000 training + 10,000 test

The Goal: Train a model to correctly label a handwritten digit it has never seen before

Sample digits

One example per class. Real handwriting varies in style, stroke and slant.



Data Preparation: Loading & Splitting

MNIST comes pre-split: 60,000 train + 10,000 test

We split 60,000 train set further into 80/20

- **Training Set (80%):** 48,000 images that the model is learning from
- **Validation Set (10%):** 12,000 images to prevent models from overfitting (memorizing the set)
- **Test Set (10%):** 10,000 images, final test on data the model has never seem prior

Data Preparation: Normalization

Scaling pixel values from their original range (0-255) to a new range (0-1).

Neural networks learn much faster and more reliably when their input numbers are small. It helps the training process run smoothly.

We use a division : `image / 255.0`

Model Selection: What makes a CNN a good choice?

CNN learns features automatically.

- Early layers learn simple shapes like edges and curves.
- Deeper layers combine to learn more complex features

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Model Architecture

- We'll use a simple, sequential CNN architecture.
- It's a stack of layers that progressively learns more complex features.
  1. **Convolutional Layer:** 32 filters of size 3x3 (input shape 28x28x1)
  2. **Pooling Layer:** Shrinks the image down. (2x2)
  3. **Convolutional Layer:** 64 filters of size 3x3
  4. **Pooling Layer:** Shrinks it again.
  5. **Flatten Layer:** Prepares data for the final classifier.
  6. **Dense Layer (64 units):** Output layer (10 units, softmax)

Activation Functions

- Activation functions decide which signal each neuron passes on
- ReLU (hidden layers): outputs max(0,x). Simple and fast, avoids vanishing gradient problem.
- Softmax (output layer) : converts raw scores into a probability distribution over 10 digit classes that sums to 1
- Softmax pairs naturally with cross-entropy loss

Model Compilation

Before training, we need to configure the learning process.

1. **Optimizer:** `Adam`
   - The "engine" that updates the model to make it better. Adam is an efficient, all-purpose choice.
2. **Loss Function:** `SparseCategoricalCrossentropy`
   - This measures how wrong the model's predictions are. The goal of training is to minimize this loss.
3. **Metrics:** `accuracy`
   - The metric we want to watch during training.

Model Training

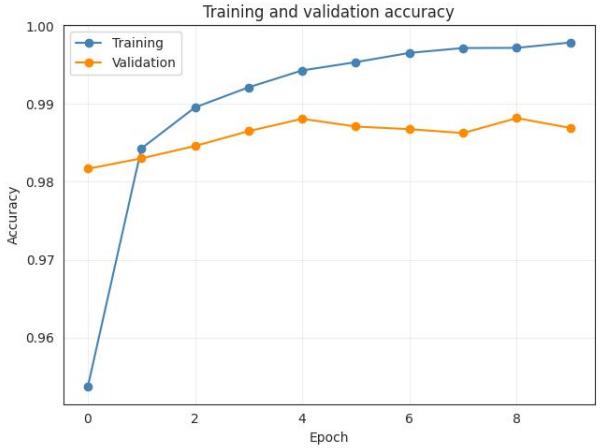
- Model is being fit to our training data.
- MNIST builds quickly, by epoch 5 validation accuracy is already close to it’s final value.
- **Epochs = 10:** The model will see the entire training dataset 10 times.
- **Batch Size:** The model looks at images in batches for efficiency purposes.
- We define **validation data** to check its performance on data it is not been training on.

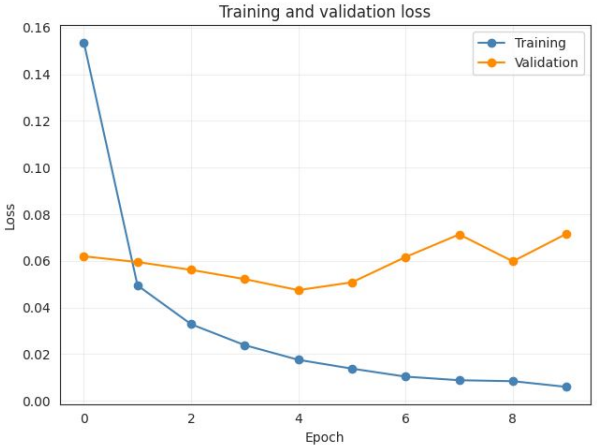
Model Performance

We can plot the training process to see how the model learned.

- **Accuracy:** Should go up for training and validation.
- **Loss:** Should go down for both.

Key Insight: After about epoch 5 validation loss began to rise slightly while training loss kept falling. Overfitting signal worth flagging





Evaluation and Testing

- We use test set - 10,000 images the model has never seen
- This gives an honest measure of generalisation
- This model achieves 99,01% accuracy (test loss: 0.0496)

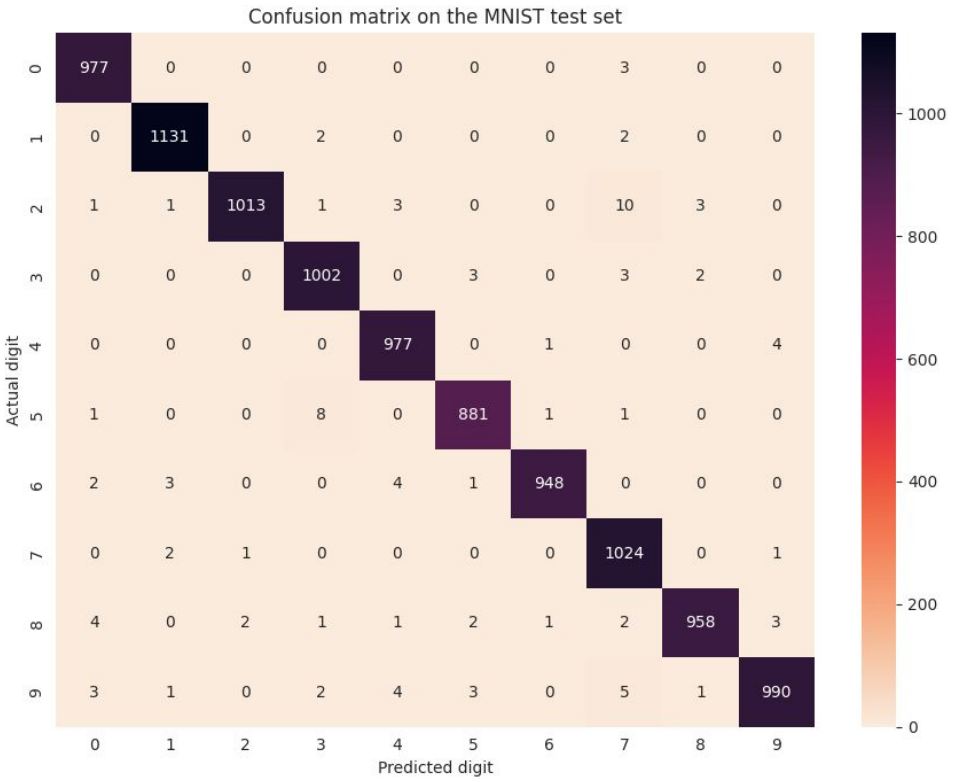
Confusion Matrix

A confusion matrix shows us where the model gets confused.

Rows: actual digit

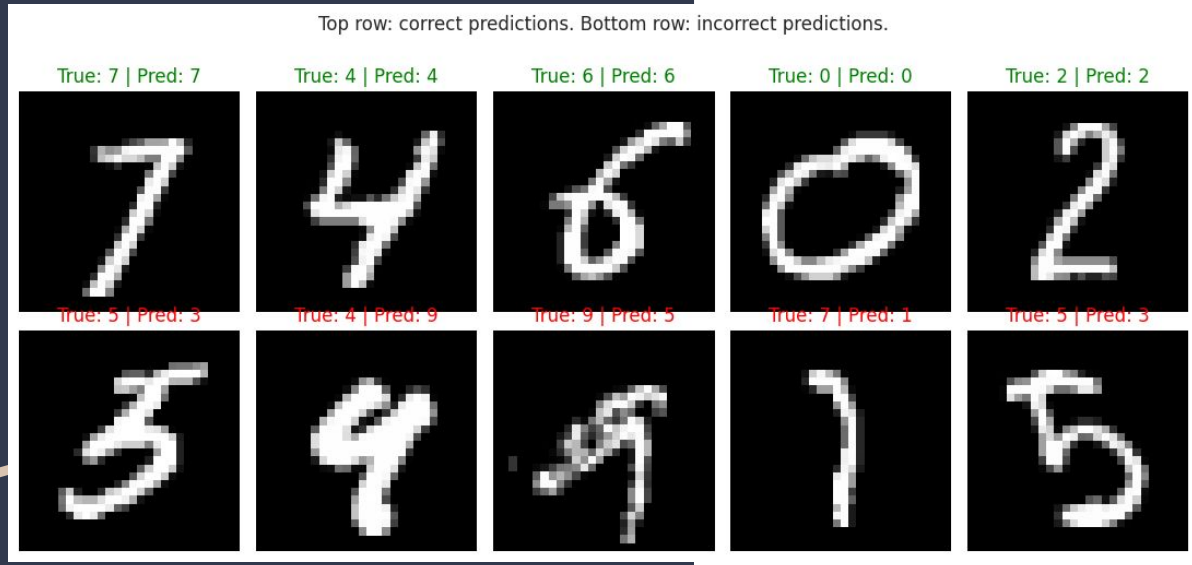
Columns: predicted digit

Off-diagonal: Mistakes are clear : we can see which digits are commonly mixed up. (2 and 7, 5 and 3, 4 and 9)



Example Predictions

- Many wrong predictions can involve ambiguous handwriting
- Suggests the model is close to the natural ceiling on the dataset



Conclusion

Conclusion: We successfully built, trained, and evaluated a CNN that can classify digits in the MNIST dataset with 99% accuracy. The project demonstrates the power of CNNs for learning features directly from images.

Lessons Learned: The importance of a proper data split (train/val/test) is critical for avoiding overfitting and trustworthy results. Even at 99% accuracy ~ 100 mistakes per 10,000 remain.

Future Improvements:

1. **Data Augmentation:** Small rotations and shifts to handle handwriting variance
2. **Regularisation:** Dropout or early stopping to address mild overfitting
3. **Harder variants**: Fashion-MNIST or EMNIST for more challenging tasks

References

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